

# FirstPass STUDIO

USER GUIDE

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

VERSION 1 · OCTOBER 2026 · BY PIVT

FirstPass STUDIO

● LIVE FROM PLAYHQ




USER GUIDE

# Graphics that follow the game.

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

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ABOUT THIS GUIDE

# Graphics that follow the game.

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

**5 s**
The live score, quarter and clock read from PlayHQ every 5 seconds while the game is scored.

**31**
Broadcast graphics, from the scoreboard and ladder to the sponsor wall, plus a test pattern.

**17**
Competition logo animations, and six intro and outro styles.

**1**
Operator on a followed night. Sequences run the run-up, the breaks and full time.

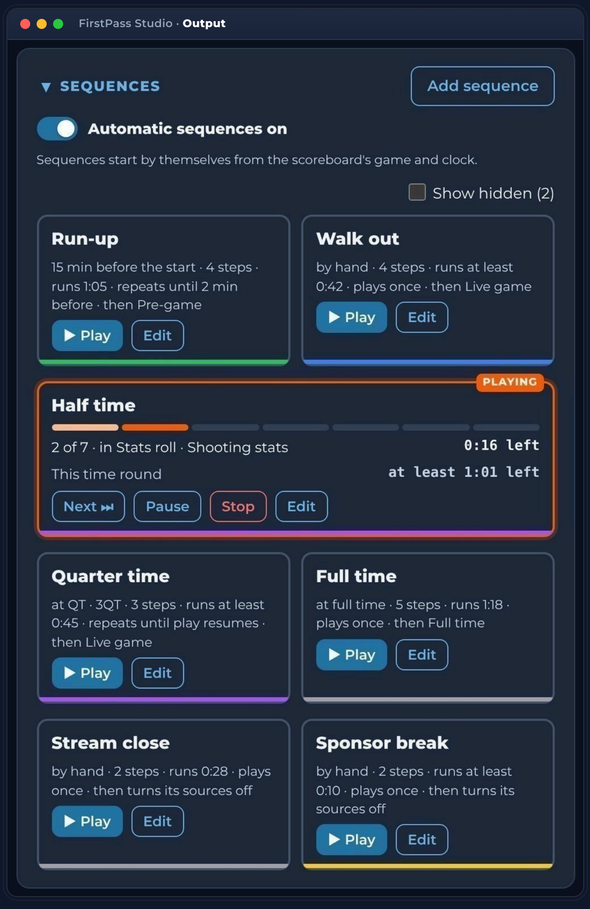
IN
### PlayHQ
The fixture, ladder, line-ups and the live score from the scorer's tablet. No account, no key.

→

ON YOUR MAC
### FirstPass Studio
Builds one 1920 × 1080 scene from layers: the base scene, automatic layers, scenes and sequences.

→

OUT
### Your switcher
A display, a Blackmagic device or NDI, with fill and key or a chroma colour over the cameras.







The Output page runs the show; the super shot plays by itself.

INSIDE


01 **What Studio is** — The app, the words used and the pages you work in.

02 **Getting started** — Install, on air and off air, the 1P menu, the checklist.

03 **Game night, the automated way** — Follow PlayHQ, the base scene, automatic layers, sequences, music, output.

04 **Running it by hand** — Manual scoring and clock, and a two-person crew.

05 **The graphics** — Every graphic, pictured, with the options worth knowing.

06 **Competitions setup** — PlayHQ link, rules, finals, colours, logos, live viewers.

07 **Sponsors** — Tiers, spots, ads, sponsored moments and proof of play.

08 **Athletes** — Photos, silhouettes, positions and games played.

09 **Stream Deck** — The plugin's keys, the links and timed keys.

10 **Settings and care** — Data folder, iCloud, backups, updates and fixes.

11 **Quick reference** — Keyboard, the night in nine lines, where things are set.

**The sample league** — Every screenshot comes from Studio running the fictional Elite Netball League: eight clubs named after animals, three divisions, a live game and fictional sponsors.

CHAPTER 1 OF 11

# What Studio is

01

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

1 The words used in this guide
2 The pages you work in
3 The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

## 1.1 The words used in this guide

### *Studio*
The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.

### *On air, off air*
Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.

### *A graphic, or a page*
One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.

### *A source*
A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.

### *The scene*
Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.

### *A scene (on the Scenes list)*
A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.

### *A sequence*
A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.

### *The competition*
Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.

### *The data folder*
Where Studio saves everything, outside the app, so updates never touch it.

## 1.2 The pages you work in

Every Studio window has the same bar at the top, with the FirstPass Studio mark and the tabs in three groups by what you are doing. **Run** is the night itself: **Output**, where the show is built and sent out, and **Scoreboard control**, which runs the board. **Plan** is before the night: the **Production Planner** plans the court and the commentators, and **Socials** makes the night's pictures for social media. **Set up** is done once: **Recordings**, **Competitions setup**, **Sponsors**, **Athletes** and **Devices**. **Docs** sits on its own at the right: it is this guide, with the bar on top. In the app, ⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices and Docs.

The right of the bar carries the status chips, on every page, so you can see the night from wherever you are: **ON AIR** in orange like a tally light, with a dot per output that is solid in the output's colour while it sends and hollow while it doesn't (or **Off air**); the game on the board, its dot orange before the clock has started, green from the first clock start or score until an hour after the game ends, then grey; PlayHQ, whose dot is the link alone (green when PlayHQ answers, amber when a reading is overdue, red when it keeps failing, grey when scores are by hand) and whose words give the quarter and clock, or **game completed** once PlayHQ closes the game (**full time** when the last quarter is run down by hand); the sequence playing with its step and the time left on it, or the triggered sequence counting down to start; **Run order** ▶ with what Play next plays and ⏭ with what follows it (also what Skip leaves at the head), shown whatever is playing and gone when the run order is empty; **ATEM** while a switcher is set up (green when it answers, red when it doesn't), and **REC** while a game records (grey while Record games waits for one). The chips are status, not links; the tabs take you to a page.

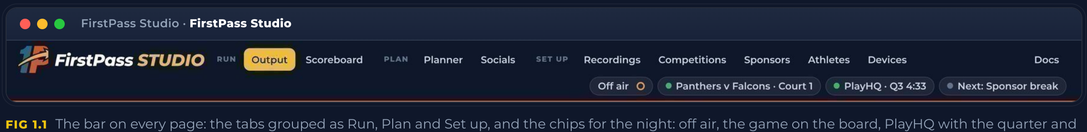
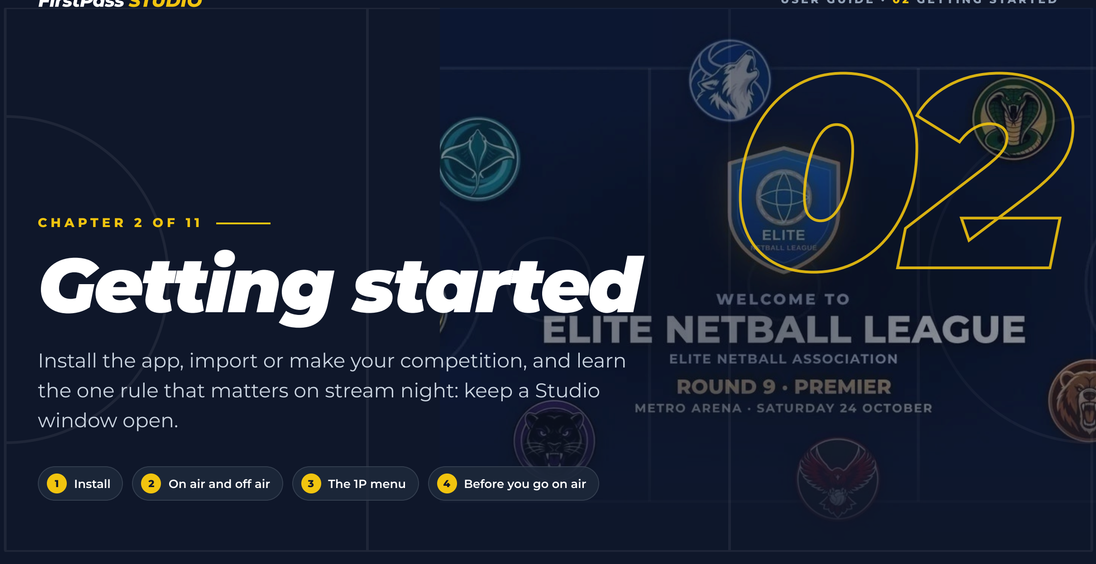



**FIG 1.1** The bar on every page: the tabs grouped as Run, Plan and Set up, and the chips for the night: off air, the game on the board, PlayHQ with the quarter and clock, and what the run order plays next.

> ***Studio does one job on stream night: graphics.***
>
> CHAPTER 1 · WHAT STUDIO IS

## 1.3 The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.




CHAPTER 2 OF 11

# Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

1 Install
2 On air and off air
3 The 1P menu
4 Before you go on air

## 2.1 Install

### 01

Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).

### 02

The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.

### 03

Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).

### 04

Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

## 2.2 On air and off air

The pages zoom like Safari's, for a small screen: **Zoom In** (⌘+), **Zoom Out** (⌘−) and **Actual Size** (⌘⌥0) in the View menu, which shows the level, or a pinch on the trackpad. Every Studio window zooms together and the level is kept on that Mac; while it isn't 100% a chip at the right of the bar shows it, and a click on the chip puts it back. The Docs tab, this guide, has its contents at the left on a wide window.

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The bar's chips show the game on the board, whether PlayHQ is answering (amber when overdue, red after a minute of errors) and which outputs are on air, on every page; the Output page's status board shows the score and the rest of the night. If anything is wrong, the page you keep open says so.

## 2.3 The 1P menu

The 1P icon in the menu bar is Studio's home: it opens Studio, its pages and this guide, backs the settings up and restores them, and has **Open at login**, **Appearance** (the Studio pages follow the Mac's light or dark look unless pinned here; the graphics never change) and **Check for updates**. Chapter 10 lists every item. Updates download in the background and install when the app quits, never mid-stream.

**LINKS** Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/`, so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

> ***The one rule that matters on stream night: keep a Studio window open.***

CHAPTER 2 · GETTING STARTED

## 2.4 Before you go on air

### 01

**Open the app.** The Output page opens and Studio goes on air. Keep a window open all night.

### 02

**Right competition.** Competitions setup shows the one in use at the top. Check its PlayHQ link and the scoring window, and that the Production Planner has tonight's court.

### 03

**Right game.** On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.

### 04

**Scores arrive.** Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.

### 05

**Scene and output.** On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.

### 06

**Sound.** Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.

### 07

**Stream Deck.** Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.

### 08

**Sponsors.** Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

# Game night, the automated way

Pipeline Studios' Automation Team

Automation Team, Core Framework, Tools, Gameplay, Audio, Animation, Lighting, Cinematics, Level Design, Narrative, Quality Assurance, Production Management, Studio Operations

## How it fits together

**Event Bus** is the backbone of the automation framework. Every system in the game can publish events to it and subscribe to events published by others, without knowing anything about the systems on the other end. A door opening, a player entering a trigger volume, a cutscene starting, a puzzle being solved: all of these are events that flow through the bus. This decoupling is what makes it possible to add new automation behaviors without touching existing code.



> *A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.*

### Boardgame control



Boardgame control

**Boardgame control** is the main window of the framework. It lists every sequence in the project, every event bus, every automation graph, and every asset that can be triggered from the outside. Designers spend most of their time here. From this window they can launch a sequence in play-in-editor mode, scrub through its timeline, inspect the state of a running graph, and see which events are currently being published on which bus.

The window is split into a few panels. The left panel is a tree view of the project's automation assets, grouped by folder and by type. The center panel shows the selected asset in detail: a timeline for sequences, a node graph for automation graphs, a property grid for event bus configurations. The right panel is a live log of events as they are published, with filters by bus, by name, and by source. Most of the debugging work happens in that log.

Boardgame control is also where you create new assets. The **New** button opens a dialog that asks for a name, a folder, and a template. Templates are important: they let a team standardize how a new sequence is set up, with the right tracks already in place and the right naming conventions applied. A new cutscene template, for example, comes with a camera track, a dialogue track, and a music track, all ready to be filled in.

Read more about Boardgame control

### The Output page



The Output page

The **Output page** is the window that shows what the framework is doing right now. It is the first place you look when something does not behave as expected. Every log message, every warning, every error raised by the automation framework ends up here, with a timestamp, a severity level, and a source. You can filter by any of those, search by text, and copy a selection to the clipboard.

The page has a toolbar at the top with a few buttons that matter more than they look. **Clear** empties the log. **Pause** stops new messages from appearing so you can read the ones already there. **Save to file** writes the current log to disk, which is useful when you want to attach it to a bug report. **Verbosity** lets you raise or lower the level of detail, from errors only up to every single event published on the bus.

Below the toolbar, the log itself is a table. Each row is one message. Clicking a row expands it to show the full payload, including the event name, the bus it was published on, and the sequence or graph that produced it. Double-clicking a row that mentions a sequence or a graph opens that asset in the Boardgame control window, so you can jump straight from a log line to the thing that caused it.

Read more about The Output page

### The boss scene



The boss scene

**The boss scene** is the sequence that plays when the player defeats the final boss of the game. It is the longest and most complex sequence in the project, and it is the one that ties together almost every system we have built. It starts with a slow-motion camera shot, hands control to the boss's death animation, triggers a particle explosion, fades the music out, and finally plays the victory stinger and the credits roll.

The scene is built as a single sequence asset, with tracks for the camera, the boss, the player, the lighting, the audio, and the UI. Each track is driven by a different part of the framework. The camera track uses the cinematic camera rig, the boss track uses the animation blueprint, and the audio track uses the music system. The sequence itself only knows about events: it publishes a "boss defeated" event at the start, and the other systems react to it on their own.

Because the scene is so long, it is split into three **shots**, each with its own camera cut. The director can preview any shot on its own, or play the whole scene from start to finish. Shots can be reordered, duplicated, or muted without breaking the rest of the sequence. This is the same workflow we use for every cutscene in the game, just on a bigger scale.

Read more about The boss scene

### Automatic layers



Automatic layers

**Automatic layers** are the part of the framework that keeps the visuals in sync with the gameplay without anyone having to think about it. Every time a sequence starts, the framework looks at the tracks in that sequence and creates the matching layers in the scene. When the sequence ends, those layers are removed. The artist never has to open the layer panel and add a new layer by hand.

The rules are simple. A track named "Fog" produces a fog layer. A track named "Rain" produces a rain layer. A track named "Lights" produces a group of lights. The framework matches track names against a table of rules that the lighting team maintains, and each rule says what kind of layer to create and what default settings it should have. If a track name does not match any rule, it is ignored, so you can add tracks for your own purposes without worrying about them.

Automatic layers are also **reversible**. If a sequence is scrubbed backwards, the layers it created are destroyed and recreated as the playhead moves. If a sequence is muted, its layers are hidden. This makes it safe to experiment with the timeline: you can always undo, and the scene will come back to exactly the state it was in before.

Read more about Automatic layers

### Scenes



Scenes

**Scenes** are the top-level container for everything that is loaded at once in the game. A scene holds the level geometry, the lights, the actors, and the sequences that belong to that part of the world. When the player moves from one area to the next, the framework unloads the current scene and loads the next one, streaming in the assets that are needed and releasing the ones that are not.

Each scene has a **manifest** that lists its dependencies. The manifest is generated automatically when the scene is saved, and it is used by the streaming system to know what to load ahead of time. If a scene is missing a dependency, the manifest will flag it as a warning in the editor, so the problem is caught before the build goes out.

Scenes can be **nested**. A large scene can be split into sub-scenes, each one covering a region of the world, and the framework will load and unload them independently based on the player's position. This is how we keep memory usage under control in the open areas of the game.

Read more about Scenes

### Sequences



Sequences

**Sequences** are the timeline-based assets that drive most of the scripted moments in the game. A sequence has a duration, a set of tracks, and a playhead that moves forward in time. Each track controls one thing: a camera, a character, a piece of audio, a UI element, or a gameplay event. The sequence editor lets you add tracks, move keyframes, and preview the result in real time.

Sequences can be **nested** inside other sequences. A cutscene sequence might contain a dialogue sequence, a camera sequence, and a music sequence, each one edited on its own and then combined. This is how we keep the work manageable: small pieces that can be reused, instead of one huge timeline that nobody wants to touch.

Sequences can also be **triggered** by events. A sequence can start when the player enters a volume, when a door opens, when a puzzle is solved, or when another sequence finishes. The trigger is configured in the sequence asset, not in code, so designers can change it without asking for help.

Read more about Sequences

### Mods, playlists and mics



Mods, playlists and mics

**Mods** are the framework's way of letting players extend the game without touching the base content. A mod is a folder that contains its own sequences, its own audio, and its own data, and the framework loads it on top of the base game at startup. Mods can override existing assets or add new ones, and they can be enabled or disabled from the options menu.

**Playlists** are ordered lists of sequences that play one after the other. They are used for the menu music, for the ambient soundscapes, and for the credits. A playlist can be looped, shuffled, or played once, and it can be paused and resumed from anywhere in the game.

**Mics** are the microphone inputs that the framework exposes to the audio system. They are used for voice chat, for in-game recording, and for the karaoke mode. The framework handles the device selection, the gain, and the noise gate, so the audio team only has to worry about what to do with the signal.

Read more about Mods, playlists and mics

### Video output



Video output

**Video output** is the framework's interface to the graphics card. It handles the swap chain, the resolution, the frame rate, and the vertical sync. It also exposes the raw frame buffer to the capture system, which is used for screenshots, for replays, and for the streaming integration.

Read more about Video output



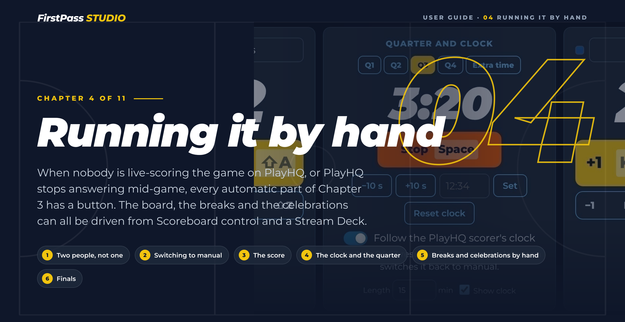




CHAPTER 4 OF 11

# Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.




1 Two people, not one · 2 Switching to manual · 3 The score · 4 The clock and the quarter · 5 Breaks and celebrations by hand · 6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

## 4.1 Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

### Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

### Production manager

- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

> ***If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks.***
>
> CHAPTER 4 · RUNNING IT BY HAND

## 4.2 Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again.

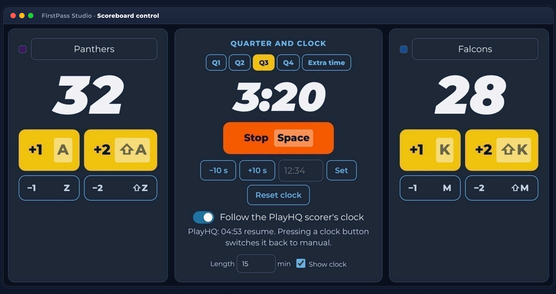



**FIG 4.1** Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away, Shift for a super shot, Space for the clock.

**WHEN PLAYHQ DROPS OUT** If PlayHQ stops answering during a followed game, the scoreboard keeps its last score, the ladder, fixtures and game list keep what they last read, and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

## 4.3 The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

| KEY | WHAT IT DOES |
|---|---|
| **Home +1 / Away +1** | A goal. |
| **Home +2 / Away +2** | A super shot: two points and the celebration. |
| **−1, −2** | Take a goal or a super shot back, quietly. |
| **Reset scores to 0** | Both scores to zero, for the next game. |
| **Super shot (home / away)** | The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1. |

## 4.4 The clock and the quarter

**Start** and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length; a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.




**FIG 4.2** Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

## 4.5 Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays: home** or **away**, or a Hot streaks source set to one streak, plays once each time it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.
- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

## 4.6 Finals




**FIG 4.3** Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.

CHAPTER 5 OF 11

# The graphics

Every great Court Sider knows, graphics are what it's all about. Each one is unique and tailored to the game. Add pages to your output and options appear as fields show up automatically. Fill in the options named here and the ones worth knowing.




- Always on
- Before the game
- During play
- After the game
- The competition
- The breaks
- Checking the outputs

All of them follow the same path: game and court picked on Scoreboard control, lists and types from its setup, and fit-long team names. Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits. Every page, with its lists, its options and a Preview button, is also listed under **Links** in the Sections menu.

## 5.1 Always on




**Scoreboard** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Scoreboard** page (PlayHQ's name for it), or the one typed as Scoreboard control, both teams with logos and colours, the score, the period and the clock. Optional: Show period, Show clock, Options position (bottom, top), Scoreboard control (a court, a run-on, or a custom one), Run on off air and the name they sign operators use.




**Score ticker** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Score ticker**. The scores from tonight's other games scrolling along the bottom, leaving out the game on the court you picked. Speed from the control page or the Settings menu, and you can hide it for a clean run-to-clip.




**Sponsor** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Sponsor** page. Title, sponsor name and optional line 2, layout on the top, bottom or alternating corners, and a scaling right-hand column. It sits up for the end of the night.




**Date** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Date**. The evening's date in the corner. Pick the format PlayHQ and Scoreboard likely to type right. Shows your control's court, or a court you name. Optional: **Show year** (hidden if it's this year) and **Show day** (add the name of the night).




**Viewers** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Watching** page. The evening's live viewer count, from the app and Scoreboard likely to type right. Shows your control's court, or a court you name. Optional: **Show peak** (adds the night's peak to the end of the night).

## 5.2 Before the game




**Countdown** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Countdown**. It counts down to the scheduled start time, and switches to the live scores. Options: heading, the words at zero, a fade over the last seconds, position and a sponsor.




**Next up** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Next up** page. It lists what's on this court for the rest of the night, with the times and teams. Options: heading, the words at zero, a fade over the last seconds, position and a sponsor.




**Title card** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Title card**. The round and game on this court for the night, with the two teams and logos. After the last game it becomes the end-of-night card.




**Match up** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Match up** page. Both sides by rude with their last five results, and the season's head-to-head. **Form** and **Head to head** can each be hidden.




**Position match ups** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Position match ups**. Each player's position, season stats, and head-to-head, arranged up the middle of the screen so it shows on one. Add one record, see one.




**Position match ups, ALL** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Position match ups, ALL**. All seven pages cut and played out as one, in court order. Options: heading, the words at zero, a fade over the last seconds, position and a sponsor.




**Team list** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Team list**. The squad of the up-first Panthers and Falcons, numbers and names, up before the game as it shows on one. Add one record, see one.




**Intro and outro** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Intro and outro** page. Plays once when it's brought up, with the sponsor's name and a tagline message. The Options style, opens from a distance, or fades in. Optional: **Outro message** for the ending, with its own style. Leave it up for the end of the night.

## 5.3 During play




**Super shot** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Super shot**. Animated, it ascends, the scoring team's logo and name, and the player who scored. Options: **Show last thought** to display **Player's name's** away / **to team** / leave a blank.




**Heating up** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Heating up** page. Shows when a team has scored a run of goals, and the number in a row. Options: **Minimum run** (the smallest run that will show), **Show streaks and rekindled** (also show when a run is broken and then restarted), **Run on off air** (how long the name stays on the screen after it's gone).




**On the siren** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **On the siren**, shown in the last seconds of a quarter. Harassers on court, and the crowd, will start cheering when you play this. Also sets the sponsor's "brought to you by" tag.




**Last 10 goals** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Last 10 goals**. Each team's last ten goals, in order, by quarter. Options: **Show quarter breaks** (mark where each quarter starts), **Show scorers** (add the player's name to each goal), and **Show only one team** (hide the other).




**Player lower third, big** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Sienna Blake, #Blake**. Name, number, position, season stats, and optional photo. Options: **Stat** (which stat to show), **Photo** (a folder of headshots, named by number, e.g. `7.png` for #7).




**Commentators, one** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Sam Lee, #Lee**. Name, role, and optional photo. Options: **Role** (their job, e.g. Commentator), **Photo** (a folder of headshots, named by number, e.g. `7.png` for #7).




**Commentators, duo** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Commentators, duo**. The same name-and-role layout for the two of them, with optional photos. Options: **Photo** (a folder of headshots, named by number, e.g. `7.png` for #7).




**Commentators, lower t...** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Commentators, lower thirds** to put their names in the corner. Options: **Show photos** (add a headshot to each), **Photo** (a folder of headshots, named by number, e.g. `7.png` for #7), **Show the scoreboard's width** (fit it to battery type), choose the Positions.




**Score worm** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Score worm**, the margin through the game as it flies, shown in case after replays. Options: **Worm** (how it's built, colours at the scoreboard's).

## 5.4 After the game




**Winner** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Winner**. The winning team, big score and a big send, washed out from the board each time it comes on. Options: **Show a result**, the words in the corner, the score and the clock.




**Full time** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Full time**. The final scores from the end of each game by the scoreboard, and the winner. Options: **Heading** (type it, or it'll fill in from the scoreboard), **Show the scores** (hide them).




**Game stats** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Game stats**. The stats by quarter, ranked by GAME SCORE, with the winner highlighted. Options: **Show quarter breaks** (mark where each quarter starts), **Show only one team** (hide the other).




**Score worm** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Score worm**, the margin through the game as it flies, shown in case after replays. Options: **Worm** (how it's built, colours at the scoreboard's).




**End of night** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **End of night**. Shown after the last game, thanks for watching, and a call to tune in next week. Options: **Heading** (the words at the top), **Sponsor** (whether it's the Profile file's sponsor, or off).

> **Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits.**
>
> CHAPTER 5 · THE GRAPHICS

## 5.5 The competition




**Ladder** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Ladder**. The division's ladder with the form pips and a marked and a top. Options: **Division** (the name on the left), **Form** (how many pips).




**Round scores** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Round scores**. The results games, or one of 10 years with **Round** set to the round you want. Options: **Round** (which round to show), **Show only one team** (hide all but one team's results), each game's division.




**Future rounds (full page)** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Future rounds (full page)**. The ladder, some fixtures and recent results. Options: **Fixtures** (how many fixtures to show), **Results** (how many results to show), **Tuesday Night Premier** (the division's name).




**Results** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Results**. The round's most recent live games are final, with the winner highlighted. Options: **Round** (which round to show).




**Mini's table** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Mini's table**. The ladder, trimmed to the competition's colours around the team you name. Options: **Mini's team** (the team to centre on), **Mini's size** (how many rows to show).




**Competition ov...** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Competition ov...**, a competition's logo, name, dates, and a grade list in their website, opening bump for Ten Options.




**Stages** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Stages**. The stages of the season's competition's colours across the top, with the current one highlighted. Options: **Stages** (which stages to show), **Show dates** (add dates to each stage), **Show only one team** (hide all but one team's competitions, then only on off for other side. Options: bold, height, type on the left, and the name of the competition and then a path applied to the text.

## 5.6 The breaks




**Sponsor card** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Sponsor card**. The sponsor's logo, tagline, offer, and a QR code to their website, lasting bump for Ten Options and the name they'll read out. Add one record, see one.




**Pivt partners** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Pivt partners** page. The name, tagline and QR of each partner, in the order you're working and the way and brand to add to the page. Add one record, see one.

**Sponsor card** · Court 1 · 1 of 1 · 1 of 1 · CLOSE UP · ON SCREEN

Pick the **Sponsor card**. It's a sponsor's logo, tagline, offer, and a QR code to their website, lasting bump for Ten Options and the name they'll read out. Add one record, see one.

## 5.7 Checking the outputs




**Test sections** · Court 1 · 1 of 1 · 1 of 1 · FULL FRAME · TAKE 1 · ONE

Pick the **Test sections**. Plays in time, names and shift shapes, and the name of each section, including each by way and to way and to way and to way and to way.

CHAPTER 6 OF 11

# Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.




Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.




A new competition starts from scratch with Studio's default settings. To start from an existing one, import its template file and change it.
A template file is a competition saved with **Export as template**, with its settings, logos and sponsors in one `.json` file.
Saved on this computer in ~/Library/Application Support/FirstPass Studio

FIG 6.1 The competitions list. Start from a template file (the repository has Thunder League, AMHNA Nationals, GIG WANs, and West Coast Thunder) or press New competition for Studio's defaults.

## 6.1 PlayHQ

Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.




Open the organisation that runs the competition on playhq.com and copy the address from the browser. For a carnival like the AMHNA Nationals, or GIG WANs, the WA Netball League, that's the organisation the competition is listed under.

FIG 6.2 The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

## 6.2 Names on the graphics

The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.




FIG 6.3 Names on the graphics.

## 6.3 Game rules

Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.




Halves give Scoreboard control HT and HZ buttons and the shooting stats say "Half time" and "Full time". Without super shots the Super columns are left off the shooting stats and top shooters (Scoreboard control keeps its +2 and -2 buttons). Every division follows the tick box unless it has its own setting below. A setting saved for a word rather than a whole division name (GIG WANs' names with "Open", "2" and "Men") covers each division with that word in its name. "Open" covers "Division 1 Open". "Men" covers "Men's" but not "Womens". A division's own setting wins over a word. When the carnival clock gets to the last minutes (3 unless you change it), the scoreboard shows a small "Super shot" tab above its clock for the divisions with super shots, until the quarter ends. It leaves the tab off. The break lengths help when building a break sequence on the Output page: its editor says how many times one loop fits in the break. With halves, only half-time is used.

FIG 6.4 Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

## 6.4 Finals

Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals names are used when it has them; otherwise the winners are picked from an Scoreboard control. **The winners are called** picks Champions or Premiers for the Social tab's picture of the grand final winners.




The 1-v-2 winner goes straight to the grand final and the loser gets another chance. The 3-v-4 loser is out. Then a preliminary final. Weeks: Semi finals, Preliminary final, Grand final

Every division uses the competition's format unless it has its own. The ladder marks the finals places by what each one earns (double chance or week off, elimination, playing for third) with a line under the last one, and the finals graphic shows who plays who week by week. PlayHQ's finals names are used when it has them (rounds with "final" or "play-off" in their name); otherwise pick the winners on Scoreboard control. **The winners are called** names the grand final winners on the Socials tab's Champions picture.

FIG 6.5 Finals formats, per division where they differ.

## 6.5 Game night

The **Fallback court to follow**, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead tonight's games are found. The Production Planner sets each game night's court; the fallback is only used on a night with no plan, which is mostly before the Planner is first used, since a night you haven't planned takes the night before it. Every grand final is set in the game's record automatically. Scoreboard control can change the court, pick a game by hand or pick another game night, for that night only. On a carnival day with several rounds, this round moves on as each round's first game comes up. **Running top shooters are called** sets the Socials picture's name: Automatic calls it Tournament top shooters when the fixture runs under 15 days and Season top shooters otherwise.




The Production Planner sets the court for each game night, and Scoreboard control follows it, picking each game automatically. The fallback court is set in case a night the Planner has no plan for (a night you haven't planned takes the night before's plan, so that's mostly before the Planner is first used). Choosing the court or picking a game by hand on Scoreboard control lasts for that night only. On a carnival day with several rounds, "this round" moves on to each round as its first game comes up. When electronic scoring starts (10 minutes before by default; 20 or 30 when games are further apart), automatic selection switches to that game, and the scoreboard only starts asking PlayHQ for its score from then on, whether the game was picked automatically or by hand. The Socials tab's running top shooters picture is called Tournament top shooters when the fixture runs under 15 days from its first game to its last, and Season top shooters otherwise, unless you pick one.

FIG 6.6 Game night.

## 6.6 Colours and font

The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one fat weight or a variable font) and it teases with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.




FIG 6.7 Colours and font, with the preview.

## 6.7 Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the banner and the signs, above the title with its option on, in the corner of the ladder and in the logo animation. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 2U" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash, without one the competition's colours are used.




FIG 6.8 Competition logo, light and dark mode.




FIG 6.9 Team logos and the teams on PlayHQ, grouped by club with a club colour each.

> **Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour.**
>
> CHAPTER 6 · COMPETITIONS SETUP

## 6.8 Live viewers

Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer-minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.




Where this competition streams: any number of YouTube channels and Facebook Pages. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page and the viewer-minutes in the sponsor report. Keys and tokens are kept on this Mac only: never in iCloud, templates or exports.

No streams yet.

The viewers graphic shows from 50 watching

FIG 6.10 Live viewers.

## 6.9 Templates, export and more

**Export as template** saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one .json file to import on any Studio or keep as a backup; **Import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use also switches the graphics to another.




The emoji either side of the title banner and The Clash's big word are set on each graphic on the Output page.

Export as template saves this competition's settings, team logos, competition logo and sponsors (with their logos; not athlete photos) as one `.json` file, to import on any Studio or keep as a backup. Importing adds its sponsors to the Sponsors page: a sponsor already there by the same name is shown on the new competition too, keeping its own details.

FIG 6.11 The More section: Client ID, export and delete.

CHAPTER 7 OF 11

# Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

1 Tiers · 2 A sponsor · 3 Spots and moments · 4 Proof of play

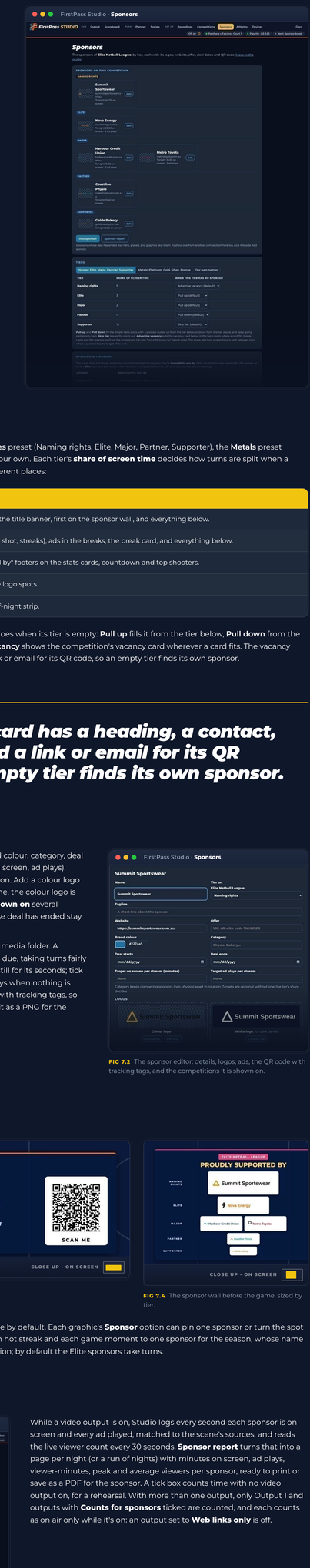






FIG 7.1

The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

## 7.1 Tiers

Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

| TIER | WHERE IT SHOWS |
|---|---|
| **Naming rights** | A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below. |
| **Elite** | The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below. |
| **Major** | Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters. |
| **Partner** | A slot in the score ticker and the logo spots. |
| **Supporter** | The sponsor wall and the end-of-night strip. |

**When this tier has no sponsor** says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

> ***The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.***
>
> CHAPTER 7 · SPONSORS

## 7.2 A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.

**Ads** are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.




FIG 7.2 The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

## 7.3 Spots and moments

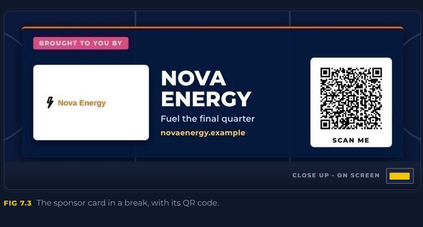



FIG 7.3 The sponsor card in a break, with its QR code.




FIG 7.4 The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

## 7.4 Proof of play

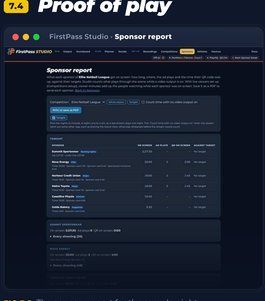



FIG 7.5 The sponsor report for the sample night.

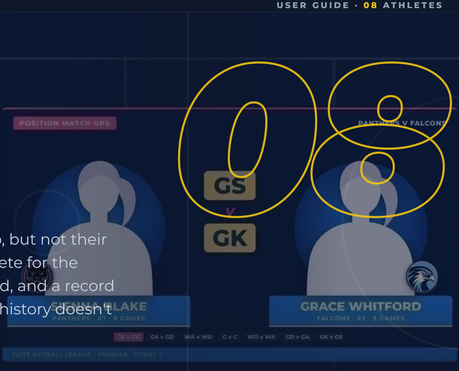
While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count every 30 seconds. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal. With more than one output, only Output 1 and outputs with **Counts for sponsors** ticked are counted, and each counts as on air only while it's on: an output set to **Web links only** is off.

CHAPTER 8 OF 11

# Athletes

PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.

1. Photos
2. Positions and games played



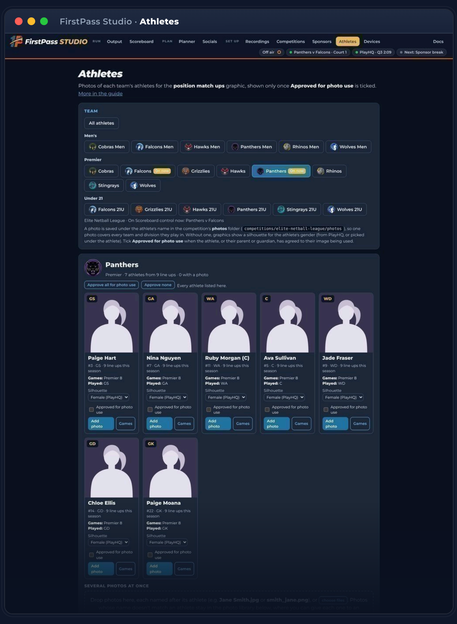




FIG 8.1

The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

## 8.1 Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Studio also remembers each athlete's PlayHQ id, so if PlayHQ changes a player's name during the season, their photo still finds them. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

A photo only goes on the stream and the Socials pictures once **Approved for photo use** is ticked under the athlete. Tick it when the athlete, or their parent or guardian, has agreed to their image being used. Until then the graphics show the silhouette. **Approve all for photo use** ticks every athlete listed for the team at once.

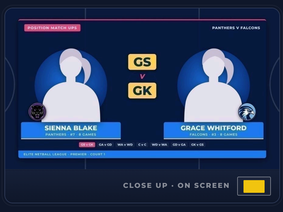



FIG 8.2 Position match ups use the photos, pairing by pairing.

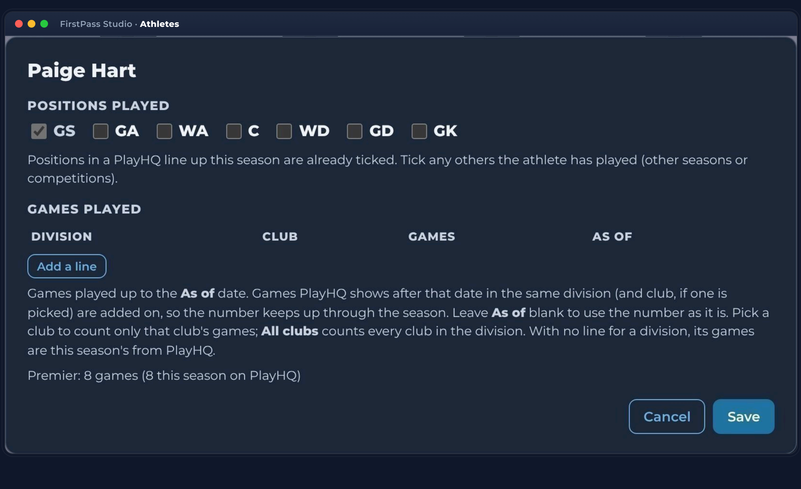



FIG 8.3 The athlete lower third, named from the register as you type on the Output page.

> ***A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo.***
>
> CHAPTER 8 · ATHLETES

## 8.2 Positions and games played

**Games** under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.



CHAPTER 9 OF 11

# Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on, plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link. It all lives on the **Devices** tab, under the list of devices allowed to change things (Stream Decks and their link sets, paired phones and programs), which is also where **Devices** in the menu bar icon goes.

1. The plugin
2. Links for everything else
3. Timed keys and the icon pack



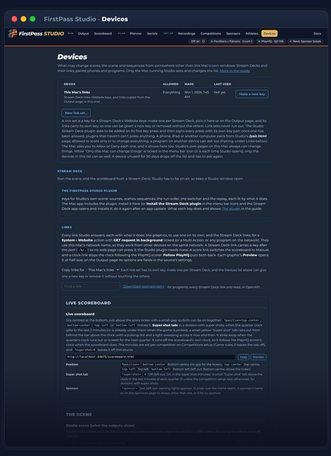



FIG 9.1

The Devices tab: the devices allowed to change things at the top, then the plugin, then Links: every link Studio answers, each with what it does, Copy and its settings.

## 9.1 The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Each kind of key has its own colour and picture, so you can see what a key does at a glance. The numbered disc in the top right corner shows which output it acts on, in that output's colour.

| KEY | LOOKS | PRESS | SHOWS |
|---|---|---|---|
| **Source** | Teal, an eye (pink with a speaker for video sound) | Show or hide (default), fade in, fade out, refresh the page or restart the video, video sound on or off | The source's name; lit while it's on, however it was turned on |
| **Scene** | Blue, stacked layers | Go to a scene, next or previous scene, or back to the current scene as saved | Lit while its scene is on, with a yellow dot when sources have changed since |
| **Sequence** | Purple, a film strip | Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing | Step and time left, like **3/5 0:12**, the whole sequence's time, both or none; **in 0:04** when one is about to start by itself |
| **Run order** | Yellow, a numbered list | Play next (default), skip what's next, load a saved order, clear the list | What plays next, like **Play next ▶ / Quarter break / +2 more**; lit while something is queued, **empty** when nothing is, a Load key names its order and how many items it holds |
| **Switcher** | Red, a camera; no disc | A camera to preview (default) or program, stinger to a camera, Cut, Auto, an upstream or downstream key, Fade to black, run a macro | The camera's name from the switcher; red on program and green on preview, a key or Fade to black red while on air, a yellow dot while the switcher isn't answering |
| **Replay** | Grey, a replay arrow; no disc | Play or pause, next or previous moment, nudge, stop the 4K replay or practice game running | What it acts on (**4K replay**, **Practice**, **Demo**); lit while one runs |

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**.

With more than one output, each key has an **Output** setting: the output it shows a source on, plays a scene on or plays a sequence on (a Switcher key has none: there is one switcher). Its disc shows the output's number in its colour (Output 1 orange, 2 pink, 3 blue, 4 green, as on the Output page), so two keys for the same source on different outputs can't be mixed up. A key for a removed output loses its disc and says **no output**.

> ***Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them.***
>
> CHAPTER 9 · STREAM DECK

## 9.2 Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Devices tab copy them, with this Mac's Stream Deck key after the port (`…:8080/k/key/deck/…`) so no web page can press them. The Studio plugin needs no key: the first time you press one of its keys the Mac asks you to **Allow** it, and from then on it signs every press with its own key pair. A link without it is refused, so copy any older Website keys' links again from Studio. With more than one Stream Deck, give each its own link set: pick **New link set…** in **Copy links for** on the Devices tab, name it after the Stream Deck, and copy its links from there. If a link gets out, choose **Devices** in the menu bar icon and press **Make a new key** on that set: its links stop working and you copy them again, while the other Stream Decks carry on. A program on another computer can ask for a key of its own: the Mac asks you to **Allow** or **Deny** it by name and address, and Devices can take that one away without touching your other keys. The Devices tab lists every one under **Links**, with what it does, the settings it takes and **Copy**; **Find** narrows the list, a graphic has **Preview** and a status read has **Try**. The useful ones:

| GROUP | LINKS |
|---|---|
| **Scores** | Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score. |
| **Scoring mode** | Follow PlayHQ, Manual, toggle. Any score key switches to Manual. |
| **Clock** | Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length. |
| **Quarter** | Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action. |
| **Game** | Automatic selection on, off, toggle; the court to follow. |
| **Ticker** | Faster, slower, normal, a set speed. |
| **Sources** | Show, hide, toggle, restart, video sound; every source out; video output on or off. |
| **Scenes** | Go to a scene, next, previous, back to the scene as saved. |
| **Sequences** | Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off. |
| **Run order** | Play what's next, skip it, clear the list, add a sequence or scene, load a saved order. |
| **Music** | Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade. |
| **Sponsors** | Play the next sponsor ad due. |
| **Switcher** | Cut, auto, stinger (to a camera or what's on preview), fade to black, a camera to preview or program, an upstream or downstream key on, off or toggled, run a macro. |

**A LAYOUT THAT WORKS** Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

## 9.3 Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.

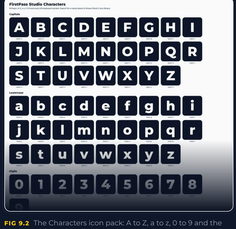



FIG 9.2 The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.




CHAPTER 10 OF 11

# Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

1 The 1P menu · 2 The data folder · 3 Settings in iCloud Drive · 4 Backup and restore · 5 Updates · 6 Links and other devices · 7 When something isn't right · 8 Data and bandwidth

## 10.1 The 1P menu

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Devices**; **User guide**; **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up...** and **Restore...**; **Open the debug log**, **Open the app log**, **Restart Studio**, **Check for updates**; **Open at login**, **Appearance** (Follow the Mac, Light, Dark), **Only this Mac can change things**, **What's new...** and Quit.

This guide comes with Studio: **User guide** in the menu bar icon, **Help > FirstPass Studio User Guide** while a Studio window is in front open it in a window of its own, and it is the **Docs** tab (⌘0) in any Studio window, with no internet needed. Phones and other computers on the network can read it too, at the Studio link with `/guide/` on the end.

## 10.2 The data folder

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

## 10.3 Settings in iCloud Drive

**Settings folder > Move to iCloud Drive...** moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac...** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac...** undoes the move. Studio goes off air for a moment for each and comes straight back.

## 10.4 Backup and restore

**Back up...** saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore...** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. This Mac keeps its own **Only this Mac can change things** choice, port and link name whatever the backup says. Make a backup after setting up a season and before trying anything new on the Output page.

## 10.5 Updates

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. If a newer version has come out since that one downloaded, it asks whether to get the newer one first or install the one it named. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

> **Updates download by themselves and install when the app quits, never mid-stream.**
>
> CHAPTER 10 · SETTINGS AND CARE

## 10.6 Links and other devices

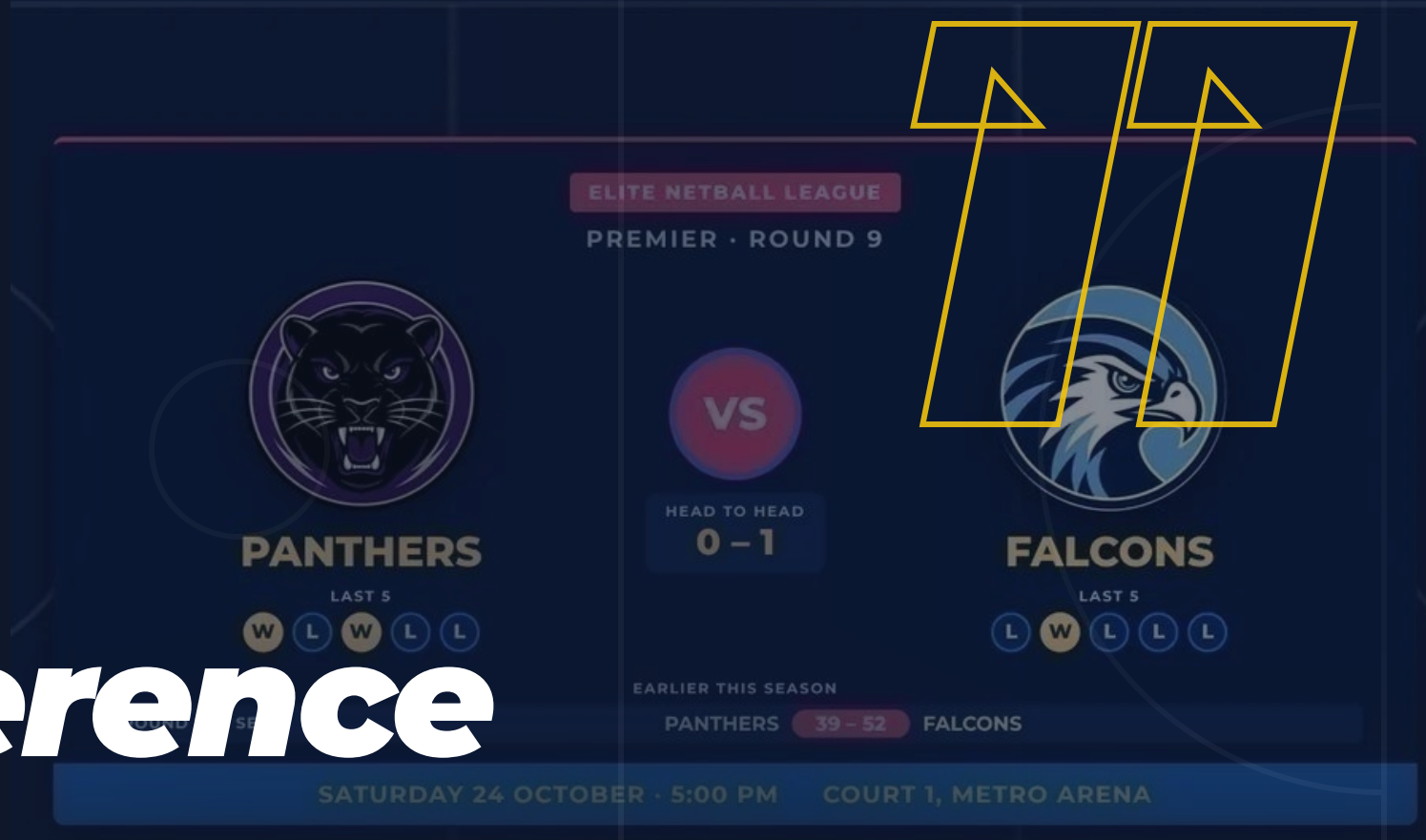
Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. Only this Mac can change scores, the scene, sequences and settings: **Only this Mac can change things** is ticked every time Studio opens. To score from a phone or let another computer drive, pair it instead: open `/pair.html` on it (the Studio link with `/pair.html` on the end), type a name and press **Ask the Mac**, then choose **Allow scoring only** or **Allow everything** on the Mac. A paired device works with the box ticked, and **Devices** in the menu bar icon lists it, changes what it may do, or removes it. Unticking the box lets every device on the network in until Studio quits; keep that for emergencies. On a network you don't control, leave it ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

## 10.7 When something isn't right

| WHAT YOU SEE | WHAT TO DO |
|---|---|
| **The board doesn't move before the game** | Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking. |
| **The board stops mid-game** | PlayHQ isn't answering. Switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back. |
| **The wrong night or game is on** | Pick the game night on Scoreboard control and pick the game; tick automatic selection again afterwards. |
| **A key says off air** | No Studio window is open. Open Studio from the 1P menu. |
| **A graphic looks stale** | Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time. |
| **Studio itself stops** | **Restart Studio** in the 1P menu brings it back with the scores and clock it had. |
| **Anything else** | **Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time. |

## 10.8 Data and bandwidth

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game (or as set, down to one a second), the ticker's other courts every 30 seconds, the fixture about once a minute for automatic selection, and the rest once when a graphic loads. The Production Planner and the Socials tab read the same fixture, results and ladders through the same cache, and finished games' line ups once, so they add nothing; the Fixture board reads other courts' live scores the ticker's way, so it adds nothing while the ticker is on and at most one request a minute per live game when it isn't. Commentator stats' auto-refresh asks Studio only; its team lists are read at the start and at each quarter break. Finished games are read once and kept. Extra outputs read the same data from Studio, so they add nothing here. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth. The PlayHQ line above the score on the Output page's status board shows it live: the ping, the bandwidth over the last minute and the data downloaded since Studio went on air.




CHAPTER 11 OF 11

# Quick reference

1 Keyboard
2 Stream night in nine lines
3 Where things are set

## 11.1 Keyboard

| WHERE | KEYS |
|---|---|
| **Scoreboard control** | A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock. |
| **Any Studio window** | ⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[ back, ⌥⌘W close every Studio window. |

## 11.2 Stream night in nine lines

**01**
Open Studio from the 1P menu and keep the Output page open all night.

**02**
Check the competition in use and the court on Competitions setup.

**03**
On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.

**04**
Turn on the video output your switcher takes and show one graphic to see it land.

**05**
Play a few seconds of music and an animation with sound.

**06**
Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.

**07**
Go to the Live game scene and let the sequences run the breaks.

**08**
Pick finals winners as they happen; turn on lower thirds when people appear.

**09**
Press Stream close, turn the output off, and send the sponsor report.

## 11.3 Where things are set

| TO CHANGE | GO TO |
|---|---|
| **Colours, font, logos, rules, finals format, court** | Competitions setup |
| **Which game, scoring mode, score, clock, finals winners** | Scoreboard control |
| **What's on screen, scenes, sequences, playlists, music, mics, outputs** | Output |
| **A graphic's options (heading, position, division, animate away)** | Output, under the source's address |
| **Sponsors, tiers, ads, moments, the report** | Sponsors |
| **Photos, positions, games played** | Athletes |
| **Each night's court, the commentators and who calls each game** | Production Planner |
| **Keys and links, and the devices allowed to change things** | Devices |
| **Pictures of a game night for social media** | Socials |
| **Backups, iCloud, updates, the debug log** | The 1P menu |

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.

# FirstPass STUDIO

MADE BY PIVT

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.